

Introduction

Research Question

This study examines how force-plate and GPS-derived performance metrics contribute to understanding athlete readiness and workload across time. Specifically, it addresses the research question: How do force-plate and GPS metrics change longitudinally, and can they be used to identify fatigue or abnormal workload? Performance monitoring is essential in athletics because athletes are continually exposed to fluctuations in mechanical load, neuromuscular readiness, and day-to-day fatigue. Capturing these changes accurately is critical for optimizing performance while minimizing injury risk (Benson et al., 2020).

Metrics Selected

To monitor athlete status effectively, this study focuses on several key performance metrics. Jump height reflects lower-body explosive power and is known to decline when the neuromuscular system is fatigued (Suarez-Arrones et al., 2020). Measures such as countermovement jump (CMJ) height and reactive strength index (RSI) are consistently associated with neuromuscular fatigue, reduced rate of force development, and decreased readiness (Anicic et al., 2023; Rabelo et al., 2022). In addition, accumulated acceleration load (AAL) captures the cumulative effects of accelerations, decelerations, impacts, and changes of direction measured via GPS or local positioning systems. Spikes in AAL are particularly relevant, as they can impose substantial neuromechanical stress that may not fully resolve with short recovery periods (Bredt et al., 2020; Delves et al., 2021). Peak propulsive force further contributes to this assessment by indicating an athlete's capacity to generate maximal force during explosive actions such as jumping, sprinting, and accelerating.

By analyzing these metrics longitudinally and ensuring consistent data collection across force-plate, GPS, and positional systems, this study aims to provide a more comprehensive view of how athletes respond to training demands over time. Tracking trends in these measures allows for improved management of workload and fatigue, supports performance optimization, and offers insights into maintaining athlete health through continuous monitoring (Benson et al., 2020).

The study focuses on five metrics that are well supported in the literature and widely used in applied sport science. From Hawkins, jump height, RSI, and peak propulsive force were selected to represent neuromuscular power and stretch–shortening cycle function. From Kinexon, accumulated acceleration load and total distance were included to quantify mechanical load and overall movement volume (Delves et al., 2021; Kupperman & Hertel, 2020). Together, these metrics provide a multidimensional framework for assessing athlete readiness.

Ultimately, the goal of this research is to determine whether consistent patterns in force-plate and GPS-derived metrics can meaningfully support athlete monitoring, inform training decisions, and enable early detection of fatigue or performance decline.

Methods

Metric Selection

Metrics were selected based on sample size and coverage across athletes to ensure sufficient data for analysis. For each metric, the number of unique athletes measured, total number of measurements, and the temporal range of data collection were assessed. Metrics recorded in at least ten distinct athletes were retained and ranked by total measurement count to prioritize those with the strongest longitudinal coverage. From this process, five metrics were selected for

analysis: jump height, peak propulsive force, reactive strength index (RSI), accumulated acceleration load, and total distance.

Data Filtering and Cleaning Approach

Performance data was extracted from a table containing athlete monitoring data from Hawkins force plates, Kinexon GPS, and Vald strength testing systems. Records with missing or invalid athlete identifiers were removed, and timestamps were standardized to define training sessions consistently. The dataset was transformed from long format to wide format to enable session-level comparisons and derived metric calculations. Data quality was assessed by quantifying missing values for each selected metric. Accumulated acceleration load had 0.25% missing records, and total distance had 1.19% missing records, while the remaining selected metrics had no missing data, indicating that there was sufficient data to address the research question. Team-level data coverage was evaluated by calculating the percentage of athletes with at least five measurements per metric. These results were visualized in a Tableau dashboard displaying team, metric, number of athletes tested, and corresponding percentages. Basketball consistently showed a high proportion of athletes meeting the minimum measurement threshold across all the selected metrics.

Statistical Methods

Standardized measures, including z-scores and percentile rankings, were calculated to compare athlete performance relative to team norms and to identify high and low-performing individuals. These standardized results were visualized using a Tableau dashboard filtered by team and metric. Across most selected metrics, men's and women's basketball consistently showed the highest-performing athletes, leading to a further analysis on these two teams. For longitudinal analysis, two athletes from the men's basketball team with the highest number of recorded

measurements were selected, and their performance trends over multiple years were analysed using time-series visualizations and simple linear regression. To address gaps in the literature regarding women's sports performance, comparative analyses between men's and women's basketball were conducted using t-tests and visualizations such as box plots.

Results

A comparative analysis was conducted between the Stony Brook Men's Basketball and Stony Brook Women's Basketball teams using five selected performance metrics derived from Hawkins force-plate testing and Kinexon tracking systems. Independent two-sample t-tests were used to evaluate differences between teams across all available observations in the dataset.

Mechanical Load and Movement Volume (Kinexon Metrics)

For AAL a statistically significant difference was observed between Men's Basketball ($n = 9,224$) and Women's Basketball ($n = 8,395$) ($t = -2.84$, $p = 0.0046$). Although statistically significant, the magnitude of this difference was small, showing that both teams experience broadly similar acceleration-based mechanical demands. This suggests similar exposure to high intensity accelerations and decelerations. This is consistent with the stop and go nature of basketball across both teams.

In contrast, Total Distance did not differ significantly between Men's Basketball ($n = 9,224$) and Women's Basketball ($n = 8,395$) ($t = 1.63$, $p = 0.104$). This finding indicates that overall movement volume during practices and competition is similar between the two teams, aligning with expectations given identical court dimensions.

Neuromuscular Performance Metrics (Hawkins Force-Plate Metrics)

Substantial and statistically significant differences were observed across all three neuromuscular metrics.

Table 1. Comparison of Hawkins Force-Plate Metrics Between Men’s and Women’s Basketball

Metric	Men’s Basketball (n)	Women’s Basketball (n)	t-Statistic	p-Value
Jump Height (m)	920	613	34.77	$< 1 \times 10^{-190}$
Peak Propulsive Force (N)	920	613	35.39	$< 1 \times 10^{-194}$
Reactive Strength Index (RSI)	917	613	18.12	$< 1 \times 10^{-65}$

Table 1 summarizes the comparison of Hawkins force-plate metrics between Men’s and Women’s Basketball. Across all three neuromuscular performance measures, Men’s Basketball demonstrated significantly higher values than Women’s Basketball. The large t-statistics and extremely small p-values indicate differences between teams, reflecting greater force production capacity, explosive power, and stretch-shortening cycle efficiency in the men’s group. These findings are consistent with established literature on sex-based differences in neuromuscular performance and support the use of force-plate metrics as sensitive indicators of athlete readiness and physical capacity in basketball populations. Our boxplot visualizations showed clear separation between teams across all three Hawkins metrics, with limited overlap in distributions.

Testing Frequency Over Time

Analysis of testing frequency over time revealed strong seasonal trends for both Men’s and Women’s Basketball. Testing volume increased during pre-season and in-season periods and declined during off-season months, particularly during summer breaks and winter academic

recesses. These patterns were consistent across multiple years, reflecting expected fluctuations in structured training and competition schedules.

Although both teams followed similar seasonal cycles, Women's Basketball displayed higher testing volumes during certain peak months, suggesting potential differences in monitoring emphasis or testing compliance rather than differences in athlete participation alone.

Department-Wide Dashboard Trends

A department-wide analysis of total tests per month across all data sources highlighted notable changes in data collection practices over time. Early data collection was primarily driven by Hawkins force-plate testing, while Kinexon emerged as the dominant data source beginning in mid-2022. From this point forward, Kinexon contributed substantially higher volumes of data, reflecting the advantages of continuous, passive tracking systems compared to scheduled force-plate testing.

Overall testing volume increased markedly during 2023–2024, coinciding with broader system adoption across teams. Despite this growth, seasonal declines in testing persisted across all systems, aligning with expected off-season and academic break periods.

Discussion

Our findings align closely with existing literature showing that CMJ height, RSI, and related force-plate variables are highly reliable indicators of neuromuscular readiness, with minimal day-to-day variation (Fitzpatrick et al., 2021; Jarvis et al., 2022; Gruber et al., 2022). Drops of $\geq 10\%$ are widely recognized as meaningful signs of fatigue because they exceed natural measurement noise, which supports our use of this threshold in the flagging system (Anicic et al., 2023; McMahon et al., 2022). A similar pattern emerged with Accumulated Acceleration Load

(AAL), where sudden spikes represent high mechanical stress from accelerations and decelerations not captured by distance alone (Delves et al., 2021). This matches research identifying acceleration-based load as a strong predictor of overuse risk, particularly in stop-and-go sports (Benson et al., 2020; Xiao et al., 2021).

Gaps in Literature & Practical Implications

Our analysis helps address a key gap in the literature: few studies integrate neuromuscular, mechanical, and volume-based workload metrics into a single monitoring framework. By combining CMJ/RSI, peak propulsive force, AAL, and total distance, our approach offers a multidimensional view of readiness rather than relying on a single indicator (Jarvis et al., 2022; Freitas et al., 2025). We also acknowledge the need to further justify and explore the sensitivity of our AAL threshold (150%), as results may shift with more conservative or liberal cutoff values, particularly given recommendations to interpret acceleration-based metrics relative to team norms rather than universal thresholds (Delves et al., 2021; Freitas et al., 2025). The system provides coaches with clear, evidence-supported rules for identifying fatigue and abnormal workload. A ten-percent CMJ/RSI drop suggests reduced readiness (Fitzpatrick et al., 2021), an AAL spike flags elevated stress (Xiao et al., 2021), and low distance highlights restricted participation or under-recovery when interpreted alongside other workload measures (Benson et al., 2020). Our team-relative performance visualizations reinforce these insights by contextualizing each athlete's outputs within their team norms, which is recommended when device-specific measurement differences exist (McMahon et al., 2022).

Our dataset revealed that Kinexon produced far more continuous data than Hawkins or VALD, mainly because passive tracking generates information automatically while force-plate tests require deliberate scheduling. This imbalance should inform future monitoring strategies, as

consistent data coverage depends heavily on how each system collects information rather than on technical validity alone (Kupperman & Hertel, 2020; Blauburger et al., 2021).

Limitations

There are several limitations that warrant acknowledgment. While interpreting the results, the team names were inconsistent and required additional cleaning and standardization before analysis. Although being necessary to ensure proper alignment of the records, this step introduces minor sources of error. Secondly, some of the metrics were unevenly available across the teams due to differences in monitoring equipment used by each sport. For instance, Kinexon GPS data was collected more consistently than Hawkins force-plate or VALD metrics, resulting in the GPS-based workload metrics having stronger coverage than the jump metrics. Therefore, this imbalance limits accurate comparisons across all metrics and teams.

Future Work

Based on these limitations, future work needs to focus on strengthening the consistency, completeness, and predictive value of the monitoring system. Through improving the standardization across teams and equipment sources to ensure that key performance and workload metrics are collected more uniformly. Allowing improved alignment in data collection practices to strengthen cross- team comparisons and enhance the interpretability of longitudinal trends.

Additionally, increasing testing frequency is the other priority, particularly during the transitional periods such as preseason and post-break intervals would further reduce gaps in the data and support more stable trend analysis. Due to more consistent data capture across the calendar year

would allow for cleaner identification of meaningful changes in performance and workload over time.

Furthermore, utilizing a larger and more consistent dataset will move the project towards predictive modeling and would further support early detection of fatigue, identifying emerging patterns of overload, and improving the ability to anticipate periods of increased injury risk.

Ultimately, these enhancements would enable the development of more proactive, data driven decision support tools to assist coaches and performance staff in managing athlete readiness and recovery.

References

- Anicic, Z., Janicijevic, D., Knezevic, O. M., Garcia-Ramos, A., Petrovic, M. R., Cabarkapa, D., & Mirkov, D. M. (2023). Assessment of Countermovement Jump: What Should We Report?. *Life (Basel, Switzerland)*, 13(1), 190. <https://doi.org/10.3390/life13010190>
- Benson, L. C., Räisänen, A. M., Volkova, V. G., Pasanen, K., & Emery, C. A. (2020). Workload a-WEAR-ness: Monitoring Workload in Team Sports With Wearable Technology. A Scoping Review. *The Journal of orthopaedic and sports physical therapy*, 50(10), 549–563. <https://doi.org/10.2519/jospt.2020.9753>
- Bishop, C., Read, P., Chavda, S., Jarvis, P., & Turner, A. (2019). Using Unilateral Strength, Power and Reactive Strength Tests to Detect the Magnitude and Direction of Asymmetry: A Test-Retest Design. *Sports*, 7(3), 58. <https://doi.org/10.3390/sports7030058>
- Blauberger, P., Marzilger, R., & Lames, M. (2021). Validation of Player and Ball Tracking with a Local Positioning System. *Sensors (Basel, Switzerland)*, 21(4), 1465. <https://doi.org/10.3390/s21041465>
- Bredt, S. D. G. T., Chagas, M. H., Peixoto, G. H., Menzel, H. J., & de Andrade, A. G. P. (2020). Understanding Player Load: Meanings and Limitations. *Journal of human kinetics*, 71, 5–9. <https://doi.org/10.2478/hukin-2019-0072>
- Delves, R. I. M., Aughey, R. J., Ball, K., & Duthie, G. M. (2021). The Quantification of Acceleration Events in Elite Team Sport: a Systematic Review. *Sports medicine - open*, 7(1), 45. <https://doi.org/10.1186/s40798-021-00332-8>

- Fitzpatrick, J. F., Hicks, K. M., Russell, M., & Hayes, P. R. (2021). The Reliability of Potential Fatigue-Monitoring Measures in Elite Youth Soccer Players. *Journal of strength and conditioning research*, 35(12), 3448–3452.
<https://doi.org/10.1519/JSC.0000000000003317>
- Freitas, J., Moreira, A., Carvalho, J., Conceição, F., & Estriga, L. (2025). Understanding Acceleration-Based Load Metrics: From Concepts to Implementation. *Sensors (Basel, Switzerland)*, 25(9), 2764. <https://doi.org/10.3390/s25092764>
- Gruber, M., Peltonen, J., Bartsch, J., & Barzyk, P. (2022). The validity and reliability of counter movement jump height measured with the Polar Vantage V2 sports watch. *Frontiers in sports and active living*, 4, 1013360. <https://doi.org/10.3389/fspor.2022.1013360>
- Jarvis, P., Turner, A., Read, P. J., & Bishop, C. (2022). Reactive strength index and its associations with measures of physical and sports performance: A systematic review with meta-analysis. *Sports Medicine*, 52(2), 301–330.
<https://doi.org/10.1007/s40279-021-01566-y>
- Kupperman, N., & Hertel, J. (2020). Global Positioning System-Derived Workload Metrics and Injury Risk in Team-Based Field Sports: A Systematic Review. *Journal of athletic training*, 55(9), 931–943. <https://doi.org/10.4085/1062-6050-473-19>
- McMahon, J. J., Ripley, N. J., & Comfort, P. (2022). Force Plate-Derived Countermovement Jump Normative Data and Benchmarks for Professional Rugby League Players. *Sensors (Basel, Switzerland)*, 22(22), 8669. <https://doi.org/10.3390/s22228669>
- Suarez-Arrones, L., Gonzalo-Skok, O., Carrasquilla, I., Asián-Clemente, J., Santalla, A.,

Lara-Lopez, P., & Núñez, F. J. (2020). Relationships between Change of Direction, Sprint, Jump, and Squat Power Performance. *Sports (Basel, Switzerland)*, 8(3), 38.

<https://doi.org/10.3390/sports8030038>

Xiao, M., Nguyen, J. N., Hwang, C. E., & Abrams, G. D. (2021). Increased Lower Extremity Injury Risk Associated With Player Load and Distance in Collegiate Women's Soccer. *Orthopaedic journal of sports medicine*, 9(10), 23259671211048248.

<https://doi.org/10.1177/23259671211048248>